

What can be claimed about a corpus of seven thousand signs: null tests, calibration and confounders in Linear A, with Etruscan as control

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Abstract

Small undeciphered corpora invite two opposite errors: readings obtained from sound alone, which any large lexicon can supply, and agnosticism that treats every statement as untestable. This paper describes a procedure that sits between the two and applies it to Linear A (7,000 signs, 1,719 documents, 799 distinct sign-groups): every distributional claim is tested against a permutation null that preserves what must be preserved; the instruments are calibrated beforehand on Etruscan, where the answers are known; the confounders of site, period, genre and scribal hand are separated where the data allow and declared inseparable where they do not; and negative results are reported with their numbers. On Linear A the procedure fixes a map of what the corpus can support: the language is not Greek even when read with Greek values; the phonotactic distance between the accounting archives and the cult inscriptions is as large as the distance to Greek and its causes cannot be separated; twenty names of the Haghia Triada archive survive in Knossian Linear B under an explicit rule; three suffixes and one prefix have measurable distributions; vocabulary sharing at Haghia Triada is explained by scribal hand rather than by document type; sound-based matching does not discriminate among candidate languages; and two-syllable form matches carry no signal in either Aegean script. The same procedure produced, and then corrected, one lexical proposal, reported elsewhere.

Keywords: Linear A, Etruscan, permutation tests, null models, calibration, confounders, undeciphered scripts, computational epigraphy.

1. The need

Linear A has some seven thousand signs distributed over 1,719 documents, most of them accounting tablets from Haghia Triada and Khania, with no bilingual and no known linguistic relative. The attempts at reading, from Gordon to Georgiev and from Palmer to Finkelberg, have followed the same path: assign to the signs the values of Linear B, search a known language for similar words, and find them, because a syllabary of ninety signs and words of two or three syllables produce matches with any large lexicon (Duhoux 1989). The recent computational study by Nepal and Perono Cacciafoco (2024) acknowledges three limitations, which are the ones this paper addresses: not filtering chance matches, not distinguishing loan from chance, and not measuring the power of a small corpus. The editors of the first collective volume devoted to the script (Petrakis and Steele 2025) add an observation taken here as the starting point: the published corpus is equivalent to that of a single scribe at Pylos, and there is no reason to suppose that a “critical point” of quantity exists beyond which results become reliable; the problem is not only size but the absence of a term of comparison.

What is proposed here is not a method of reading but a procedure for fixing the map of what can be claimed. Its components are not new: the distributional analysis is that of Duhoux and Davis, the fixed-margin permutation nulls come from community ecology (Gotelli 2000; Strona et al. 2014), calibration on a corpus with known answers is standard practice for any instrument. What had not been done in this field is to combine them with discipline and to publish the negatives with their numbers, and that is what follows.

2. Data and procedure

2.1 Corpus

GORILA transliterations (Godart and Olivier 1976-1985) in the machine-readable form of the LinearA Explorer (Hogan 2022), which preserves subscripts, site, support, archaeological context, the hand attributions of GORILA V and Younger's commentaries (2024b) with the seal data. Sign verification against the SigLA dataset (Salgarella and Castellan 2020). For Linear B, the same author's edition of the standard editions (5,889 documents, 5,597 sign-groups), with Olivier's Knossos hands. For Etruscan, the Larth corpus (Vico) with 7,139 records and the ETP corpus (429 texts with metadata). A known defect of the Explorer data, the support label of the Zakros tablets, was corrected from SigLA; one ligature in its transliteration (*304+PA-KU-PA on HT Wa 1020) turned out to be two consecutive signs according to SigLA, and is recorded here as a warning about derived transliterations.

2.2 Units and measures

The units are sign-groups (799 types, 1,117 tokens on tablets with two or more groups; 76% singletons), documents as sets of groups, and profiles of initial syllables and of consonantal skeletons. The measures are the usual ones: Jensen-Shannon distance between profiles (equation 1), Jaccard index between documents (equation 2), Mantel correlation between distance matrices (equation 3), mutual information between positions (equation 4), and the curveball algorithm for fixed-margin nulls on presence matrices (equation 5, Strona et al. 2014). Each is stated in the appendix with its source.

2.3 Nulls

Each claim carries the null that preserves what the claim does not attempt to explain: for co-occurrence between documents, permutations that preserve the number of groups per document and the number of documents per group (curveball); for morphology, permutation of the affixed syllable among all words with their lengths preserved; for positional classes, permutation of labels; for confounders, permutation within strata (type label within hand, or hand within type). The threshold was set at $p < 0.01$ with one thousand permutations for new claims and $p < 0.05$ for replications; multiple comparisons are corrected by Bonferroni when fewer than ten and by Benjamini-Hochberg when more.

2.4 Power

Before each test the minimum detectable effect with the available n was estimated. Two figures govern the rest: the floor of distance between two samples of the same corpus at 42 types is

0.198 (Jensen-Shannon on consonantal skeletons), so that no smaller difference is detectable at that size; and in Etruscan the sibilant test has 68 usable pairs, which is the order of magnitude of the positional tests in Linear A. A test whose expected effect falls below the floor is declared undecidable and is not run.

3. Calibration on Etruscan

Etruscan is the appropriate control: a read alphabet, a non-Indo-European language half understood, a corpus of the size of Linear A with metadata of city and period, and three established grammatical or sociolinguistic facts that the procedure must recover without being told.

Genitive before clan. Before the word clan, “son”, the father’s name stands in the genitive (-s, -l). The positional test, applied blind, finds the endings -s and -l in 58% of the words preceding clan (and sec, “daughter”, and puia, “wife”) against 26% in the rest of the corpus ($p = 0.002$ by permutation). The instrument sees inflection where it exists.

Syncope as change over time. The loss of internal vowels is dated around 500 BC. The syncope index is 0.28 higher in late texts than in early ones, and the effect survives control by city ($p < 0.002$ within city). The instrument sees time where it exists.

Sibilants as geographic variation. The graphic distinction of sibilants is a northern trait. The north/south ratio is 0.6 against 0.1 ($p = 0.72$ within city): the effect is one of city, not of period. The instrument does not confuse geography with time when the metadata allow them to be separated.

The calibration fixes what may be expected from the same apparatus on Linear A: it will recover morphology and stratification if they exist and if metadata allow the confounders to be separated; where they do not, it will say “undecidable”, and that answer must be accepted.

4. Results with nulls on Linear A

4.1 Not Greek, not even in its own syllabary

The Jensen-Shannon distance between the skeleton profiles of Linear A and Linear B, read with the same values, is 0.104, against a floor of 0.020 between two halves of A and 0.092 for a null in which the sign inventory of B is shuffled: A is further from Mycenaean Greek than scrambled Mycenaean Greek. A converter from alphabetic Greek into the syllabary recovers 36% of whole words and 63% of roots (47% and 68% with the digamma and labiovelar rule); the 85% threshold that identity of language would require is not reached. A screen of seven candidate languages by consonantal skeleton gives ratios from 1.46 (Hittite) to 1.68 (Hebrew) over each lexicon’s own null, with Greek at 1.47: none stands out, and the highest ratio belongs to the largest lexicon. Comparison by sound, whatever the language, does not discriminate.

4.2 Two registers, with inseparable causes

The distance between the accounting tablets of Haghia Triada (LM IB) and the cult inscriptions of the sanctuaries (MM III-LM IA) is 0.193, of the same order as the distance to Greek (0.194).

The four possible causes (genre, site, language, period) travel together in the corpus and cannot be separated: Phaistos, the only period control without a change of genre, has 42 types and lies below the floor. Within Haghia Triada, where site is constant, document types (personnel, produce, aromatics, allocation) do not differ in syllable profile above the null (Jensen-Shannon at 80 syllables, all pairs $p > 0.18$, with free permutation and with permutation within hand). The difference between archives and sanctuaries is therefore one of genre and site, not of administrative type; whether it is also one of language the corpus cannot say, and so it is stated. The variants of the libation formula are explained by site, not by grammatical agreement (p from 0.011 to 0.16 when controlling for site); the prefix ja-/a- has frequencies of 7 per thousand at Haghia Triada against 140-240 per thousand in the sanctuaries, which makes it a feature of genre.

4.3 Names that survive at Knossos

With Younger's explicit rule -U/-E $\rightarrow$ -o, twenty sign-groups from Haghia Triada of three or more syllables reappear in Knossian Linear B, against 8.1 expected under a bigram null ($p < 0.003$); at Pylos, 7 against 3.5 ($p = 0.06$); at Thebes, none; among the sanctuary words, 0 of 47. Survival is asymmetric (names, not cult vocabulary) and local (Knossos, not the mainland), which is consistent with a Mycenaeanised administration that retains Cretan onomastics. Two non-Greek theonyms of Knossos (pa-de, a-ta-na) appear in A (2 of 15, $p = 0.04$); three Pre-Greek names of B (mi-ta, sa-sa-me, ki-ri-ta₂) appear at Haghia Triada (3 of 20, $p = 0.09$), with the Greek controls at 0 of 6.

4.4 Morphology placed, not read

Three suffixes form root/derivative pairs above the shuffled-syllable null: -ja (8 pairs against 3.3, $p = 0.02$), -te (7 against 3.3, $p = 0.03$), -se (5 against 1.5, $p = 0.02$); none survives Bonferroni over eleven tests, and this is reported. The prefix ka- forms five pairs against 1.7 ($p = 0.013$) and concentrates its distribution on the roundels (17% of roundel groups against 4% of tablet groups). In the root/-ja pairs the bare form is a heading and the derived form an entry with a quantity; taken over all groups in -ja, the concentration disappears. This is grammar placed by position, not read. There is no word for "son of"; filiation, if expressed, goes by suffix.

4.5 Compartmented vocabulary, and who compartments it

At Haghia Triada the accounts with a total (ku-ro) have six monopoly words (groups that appear in no other document type; null 1.1, $p < 0.003$) and the allocations (sa-ra₂) three (null 0.4, $p = 0.007$). There is no measurable co-exclusion between frequent words in the whole corpus (zero pairs with expected ≥ 1 and observed 0), and the documents $\times$ words metacommunity with the curveball null has a single module (the grain list of HT 86 and 95). With the hand attributions of GORILA V (73 tablets, 21 hands), vocabulary sharing between tablets is explained by hand (effect 0.032, $p = 0.002$ controlling for type) and not by document type (0.025, $p = 0.21$ controlling for hand); the compartments are, at least in part, scribal portfolios. The claim must be calibrated: in the Linear B of Knossos (1,158 tablets with hand and series), vocabulary is explained by the editors' series (effect 0.020, $p < 0.001$) and not by hand, but the series is a

classification made by content, so the effect is partly tautological; the typology used here for Haghia Triada is coarse and our own, and the fair comparison (Montecchi's 2010 classes for HT, or a content-independent typology for Knossos) is pending. What stands is that the scribe is a confounder no lexical analysis of A had measured, and that its weight depends on the typology against which it is measured.

4.6 Form screens and their negative control

A screen of the whole vocabulary against twenty phytonyms attested as commodities in Mycenaean gives 34 stem matches against 15.4 expected (maximum 25 in 200 permutations); against sixty, 91 against 53.1 (maximum 84); against seventy-nine, 97 against 73.7 (maximum 103). The excess concentrates in one family and disappears as the list grows. The same screen by field (animals, vessels, minerals, lists of twenty) stays within chance in all three. Applied to Linear B, it produces 383 matches, most of them with Greek personal names unrelated to plants. Conclusion: a two-syllable match discriminates in neither script, and any reading resting on it is indefensible; what discriminates is the conjunction of stem, distribution and derivation, case by case, as argued in the paper on phytonyms.

4.7 The libation formula: slot function without reading

The eighteen non-accounting inscriptions carrying at least one anchor of the libation formula allow its template to be fixed by distribution. Four nearly invariant anchors (ja-sa-sa-ra-me, u-na-ka-na-si, i-pi-na-ma, si-ru-te) and a fixed opening (a-ta-i-*301-wa-ja, sixteen attestations, always in first position, with endings that cluster by site) enclose slots whose fillers are singletons in 32 of 32 cases (probability under the corpus singleton rate, 0.76^{32} , of the order of 10^{-4}): the slots hold the open class of names and places (Dikte, Ida), the anchors the closed class. The only inscription with logograms (SY Za 2) places the anchor u-na-ka-na-si beside oil and the preceding word beside olives, which fixes that anchor as the term of the offering without recourse to etymology. Three prefixes have measurable distributions (i-na- on place-names, ja-/a- as a genre feature, ta-na- in the opening), and one element (u-ti-nu) appears both free and attached without divider to two long words from Iouktas, so that the opening is a compound. The covariation between the ending of the opening and that of the following toponym is not systematic (-wa-ja accompanies -tu, -te, -a and -mi), which confirms that the variants are of site and not of agreement. Compared as structures with the contemporary votive formulae, the Minoan template (fixed opening, place and names, invariant term without logogram, offering term with logogram, two closings) resembles the Egyptian and the Anatolian, and not the Semitic or the Greek, which put the deity last; with eighteen fragments the first two cannot be told apart. This is as far as the corpus allows one to go in reading a sentence: the function of each position, not the value of any word.

5. Negative results of method

They are reported because they close paths that others would reopen.

A per-node structural descriptor (Omega-N, with the public code of BiomeMakers/OmegaN) applied to aligning A and B signs by their role in the co-occurrence graph recovers a mean rank

of 27.4 against 32.5 for chance ($p = 0.018$), exactly what degree alone recovers (26.8): what transfers between scripts is the degree share, and structural role encodes the language, not the sign. Within A, the descriptor has no test-retest reliability at these sizes (triadic excess between two halves of Haghia Triada +0.06 and +0.02), so any test of fine structure is undecidable; a minimum-size condition is proposed for the method's screen.

A function classifier (49 labels, 25 features of position and company) scores 0.67 to 0.73 under leave-one-out, against 0.39 to 0.48 with shuffled labels: it distinguishes the libation formula, and headings from entries, but classifies the two known commodities as names, because there are two examples of commodities. Structure predicts coarse function, not semantic class.

A neural decipherment model with and without the Omega-S regulariser on the twenty-five known words is preregistered with a prediction of zero in both arms; it has not been run.

Co-exclusion between words has no power at this size (section 4.5), and the aromatics "recipe" (equal proportions among ingredients) is undecidable with the legible quantities (0.26 of equal pairs against 0.21 for the fraction null, $p = 0.24$).

The ligature *304+PA-KU-PA of the LinearA Explorer does not exist: SigLA reads *629 followed by KU-PA. It is reported because it supported for two days an argument for a reading, and because derived transliterations require sign-by-sign verification before use as data.

6. A case: how the procedure produced, and then corrected, a reading

The procedure is best illustrated by the case in which it half failed. Searching for the sign-groups that behave as counted things (recurrence within a document with different quantities, integer quantities, absence from personnel lists), the screen isolated KU-PA (AB 81-03): repeated on both faces of ZA 11 with 1 and 3, on KH 29 with 2, on HT 110a with 1; with a family of derivatives (KA-KU-PA, KU-PA-ZU, KU-PA-JA, KU-PA-RI) that stand beside the logogram *303 on two roundels and on KH 5; and with two affixes that recur in the corpus above the null. Only then was the form compared: Greek κύπαρος, Pre-Greek according to Frisk and Beekes, Mycenaean ku-pa-ro, which glosses the cyperus logogram on KN Ga 517. The first version read the family as the Minoan word for cyperus, against Younger's reading, which took it for the name of a contributor.

Three later checks, all from the same procedure, changed the reading without touching the etymology. Sign-by-sign verification in SigLA showed that the "ligature" *304+PA-KU-PA on the nodule HT Wa 1020, one of the supports, was the logogram *629 followed by two signs. Tabulating the seals of all the Haghia Triada roundels (Younger's commentaries, SigLA) showed two sealing groups: one in which the word stands alone, and another in which the word is followed by a commodity logogram, and in that second group stand KA-KU-PA with *303 but also A-RA-TU-ME with ewes and KU-MI-NA-QE with goats, words that cannot be commodities. And the structure of KH 5 (group, logogram, quantity, with WI-SA-SA-NE before *626 and KU-PA-ZU before *303) turned out to be that of parties before commodities. The corrected reading is the one that reconciles: the stem KU-PA is the plant's word; the family in

the archives is a name formed on it, like Kyparissos or Marathon in Greek; and two further cases of the same type (KU-MI-NA-QE and cumin; KI-KI-RA-JA on the KI-KI-NA that Neumann identified with the sycamore fig in 1960) sustain the pattern.

What the case teaches about the method is what matters here. First, that the function-before-form order does not by itself protect against an error of function: recurrence with quantities, taken as the signature of a commodity, was compatible with a party allotted twice, and only a datum external to the tablet (the seals) decided it. Second, that every support must be verified in the primary source and not in a derived transliteration, however good. Third, that the correction did not destroy the result but placed it: the etymology, the only thing that did not depend on function, survived intact, and the pattern of names formed on plants is more defensible and more general than the initial reading. The paper presenting the case (Acedo 2026, under evaluation) carries that history inside it because the history is part of the evidence.

7. Discussion

The procedure does not decipher, and it is worth saying so with the figures of this paper: on 7,000 signs it produced two stems with a proposed meaning, two names decomposed, two affixes with a placed function, and the template of one formula. What it produced in greater abundance are negatives with numbers, and that is its value: each closes a path that the literature has reopened for a century. Comparison by sound does not discriminate among candidate languages (all at 1.3-1.7 over their null), neither in Linear A nor in Linear B (383 false matches under the same rule); the difference between archives and sanctuaries is not one of administrative type; co-exclusion is not measurable at 224 samples; a network descriptor is not reliable at this size; a classifier does not learn a class from two examples.

That the negatives are of size and not of method is shown by cross-calibration. The same apparatus, on Proto-Elamite (1,594 tablets, 8,282 entries), recovers in one session the classification of signs by numeral system, the institutional header and the name/commodity partition that the field established over twenty years, and finds in addition the frame of the long strings, 24 grain monopolies and 13 co-exclusion pairs under the curveball null, which in Linear A had no power. On Uruk (6,387 tablets), it recovers three of the four predictions on the inheritance of numeral systems and refutes the fourth by absence. On Eteocypriot, with 21 inscriptions, the profile test is declared without power and the positional test separates the two languages by their endings ($p < 10^{-16}$). On Iberian, with 1,919 texts and only the province as metadata, it recovers Untermann's onomastic formants (295 against 11 for the null) and the two known grapho-morphological isoglosses (the final southern sign S56, $p = 3 \cdot 10^{-9}$; the north-eastern suffix -*mí*, $p = 3 \cdot 10^{-5}$). In every corpus the instrument says the same thing: what there is, what there is not, and what cannot be said with that *n*.

The origin of the apparatus is community ecology, and the analogy deserves one sentence, no more: documents are samples, words are taxa, and the questions about co-occurrence, co-exclusion, specificity and turnover are the same as those asked of a microbial community, with the same nulls (Gotelli 2000; Strona et al. 2014) and with the same test-retest reliability

requirement that any index of structure demands (Acedo 2026b). What epigraphy adds to ecology is a confounder that in biological samples is called batch and here is called scribe: at Haghia Triada shared vocabulary is explained by the hand and not by the document type ($p = 0.002$ against 0.21), and a niche index that did not control for it would be measuring scribal portfolios. What ecology adds to epigraphy is the habit of asking, before reading, whether the corpus has the size to answer.

Where this leads is not towards reading. It is towards anchors: every Pre-Greek word whose stem appears in the tablets with the right function is one line of a synthetic bilingual, and with thirty such lines Minoan would have a vocabulary, a morphology on that vocabulary and a formula with parts, even if it were not pronounced. The procedure described here is the factory of those anchors, and the guarantee that each carries its null, its calibration and its correction when due.

Declaration of assistance

Corpus processing, permutation tests and the drafts of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

Appendix: equations

- (1) Jensen-Shannon distance: $JSD(P,Q) = \frac{1}{2} D(P\|M) + \frac{1}{2} D(Q\|M)$, $M = (P+Q)/2$, D the Kullback-Leibler divergence in base 2 (Lin 1991).
- (2) Jaccard index: $J(A,B) = |A \cap B| / |A \cup B|$ (Jaccard 1901).
- (3) Mantel correlation between distance matrices, with row-and-column permutation (Mantel 1967).
- (4) Mutual information between positions: $I(X;Y) = \sum p(x,y) \log_2 [p(x,y) / p(x)p(y)]$ (Shannon 1948).
- (5) Fixed-margin null by the curveball algorithm (Strona et al. 2014), which preserves the row and column sums of the documents $\times$ words matrix.

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